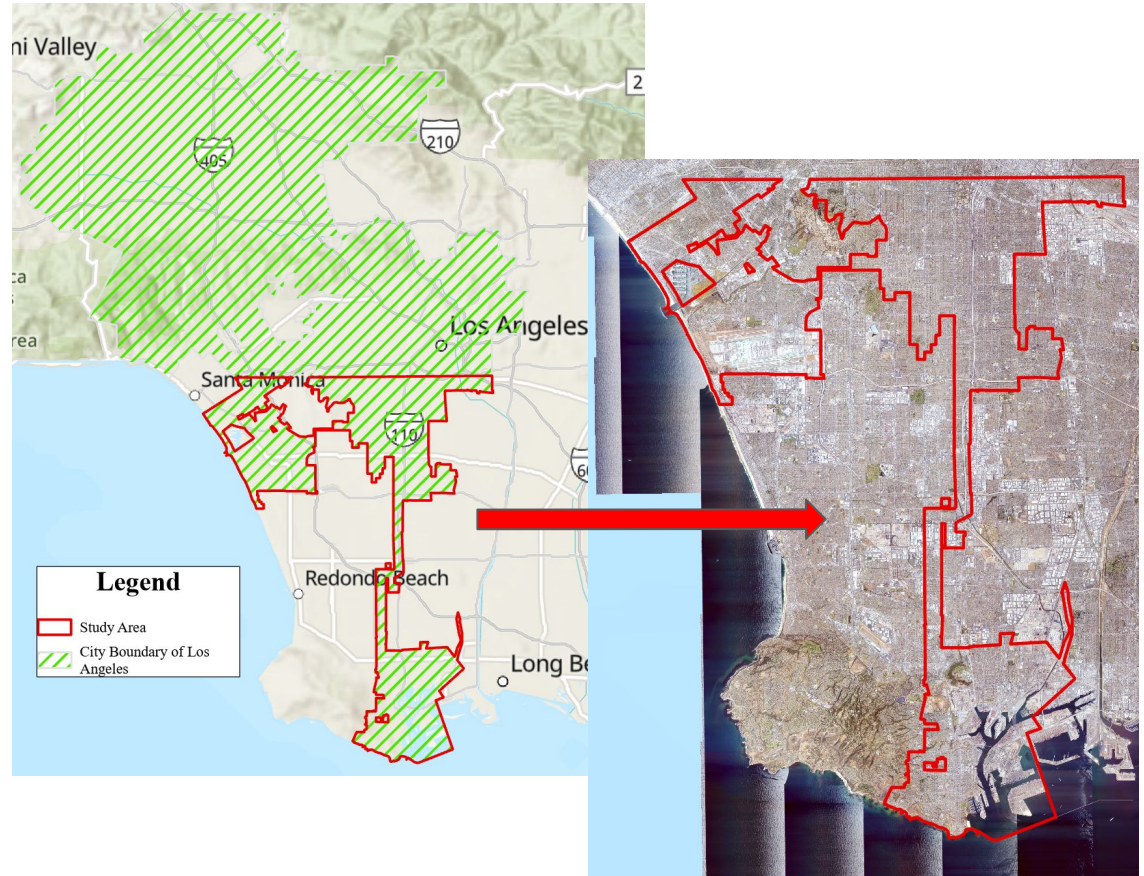


Random Forest Tree Classification

Study Area

The study area is located in the southern part of Los Angeles, specifically defined within a red boundary as depicted on the map.



Dataset

Tree Dataset	Tree location and genus table from Google Street View Dataset
	Global Canopy Height dataset from Meta https://gee-community-catalog.org/projects/meta_trees/
Image Dataset	2022 National Agriculture Imagery Program (NAIP) Aerial Imagery from USGS https://hub.arcgis.com/datasets/ca507ccde28642999d98ce0017618954/explore

Dataset - Google tree genus data

LosAngelesTrees_test ☆ 📁 ☁

File Edit View Insert Format Data Tools Extensions Help

100% | \$ % .0. .00 123 | Default... ▾

A1 ▾ | 📌 IDX

	A	B	C	D	E
1	IDX	SHAPE_LNG	SHAPE_LAT	GENUS	TAXONOMY_ID
2	61cce057-a435-	-118.2573438	33.92600997	ceratonia	64
3	8d1bc824-d49f-	-118.2569565	33.9260129	ceratonia	64
4	76774b59-ca04-	-118.2571602	33.9259182	cupaniopsis	93
5	bc4d490e-7083-	-118.2578887	33.92601225	ceratonia	64
6	f8daa689-30f0-3	-118.2588306	33.92601309	cupaniopsis	93
7	c7cd76ab-f3f1-3	-118.2593377	33.926012	cupaniopsis	93
8	762adfd9-fa1d-3	-118.2600324	33.92600955	ceratonia	64
9	6ccb649-f3bb-	-118.2597403	33.9259224	ceratonia	64
10	782c20b6-e3eb-	-118.2589665	33.92592216	cupaniopsis	93
11	1b70f9c0-142f-3	-118.259008	33.92601462	ceratonia	64
12	c74f9033-46c0-	-118.2607445	33.92561721	celtis	63
13	8e48d980-3ad4-	-118.2603942	33.92574276	jacaranda	153
14	aa336aae-6ad7-	-118.2602925	33.92587961	jacaranda	153
15	5204882e-0171-	-118.260257	33.92574888	ulmus	313
16	09642cb1-b31d-	-118.2414682	33.92965101	washingtonia	318
17	a9fe9728-efe2-3	-118.2455611	33.92962357	melaleuca	187
18	0902b09f-a014-	-118.2442989	33.92963383	callistemon	44
19	ab452ba7-d169-	-118.2447892	33.92962876	ulmus	313
20	cf03c2a1-c452-	-118.2450254	33.92962828	liquidambar	170
21	4d8a5134-85a3-	-118.2451334	33.92962729	liquidambar	170
22	1ee3d2a7-0c5c-	-118.2553858	33.93015607	cupaniopsis	93

Total 75814 trees



mapping



Create buffer
(a square with sides of
6 meters)

Dataset - NAIP data



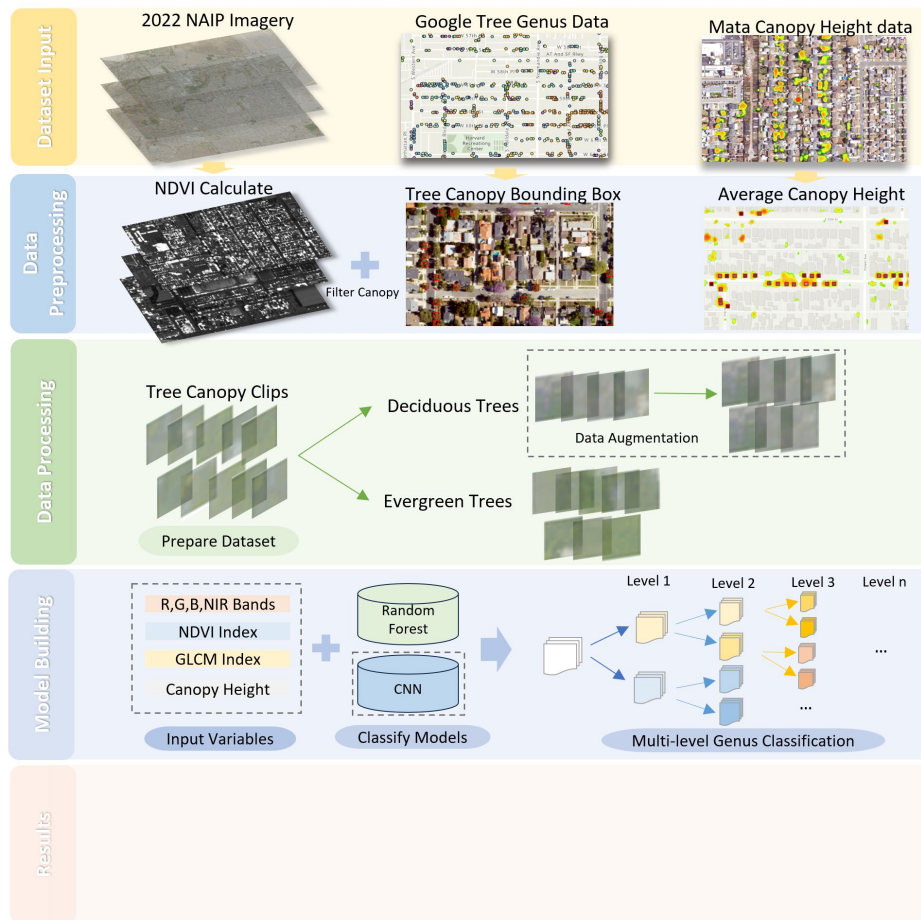
4 bands 60cm resolution



NDVI Calculate

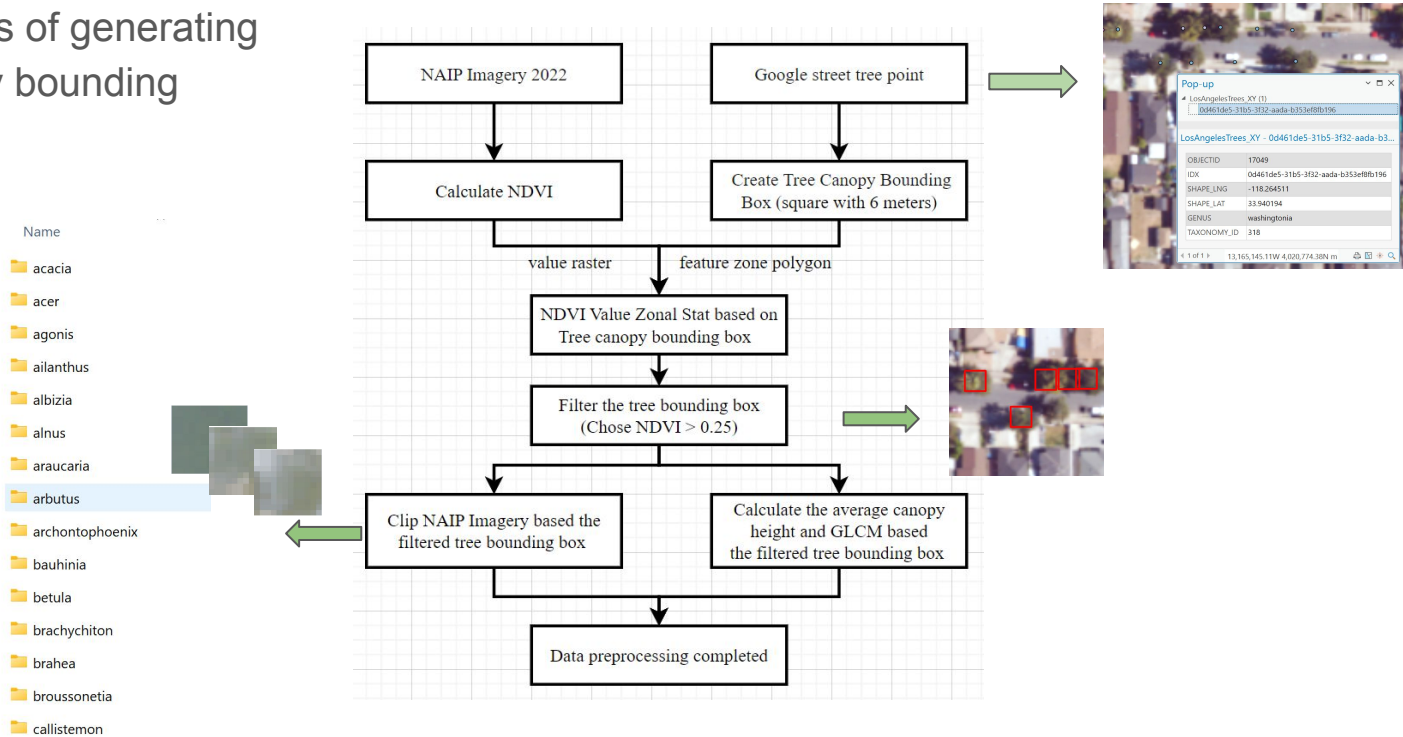
GLCM Analysis

Workflow



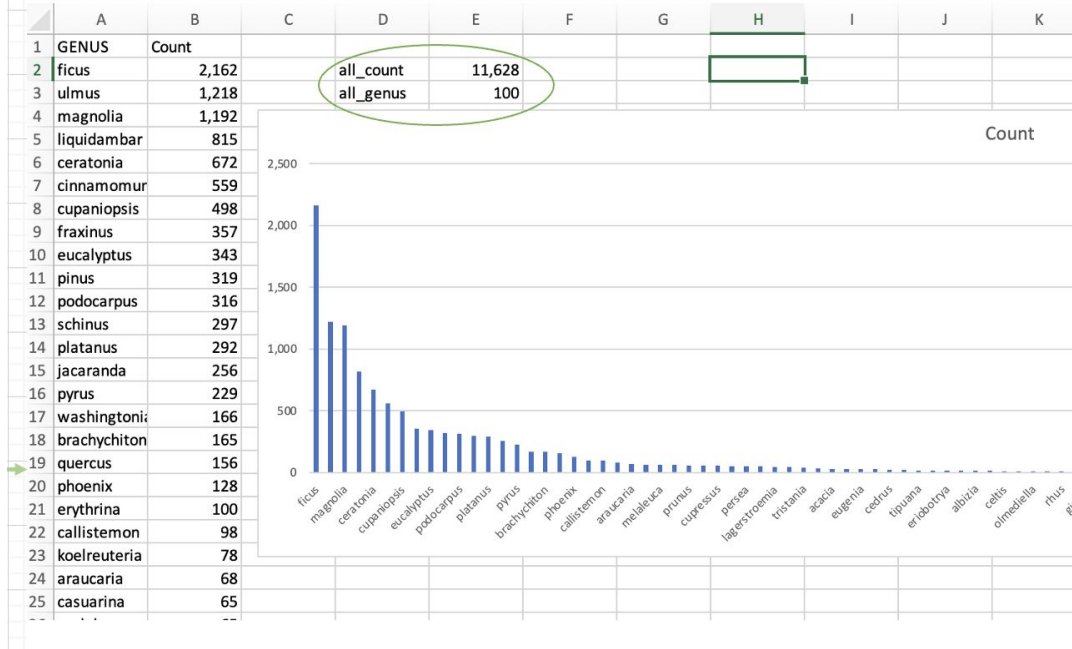
Data Preprocessing

The process of generating tree canopy bounding boxes.



Data Stat

Tree bounding boxes statics (for all NDVI>0.25)



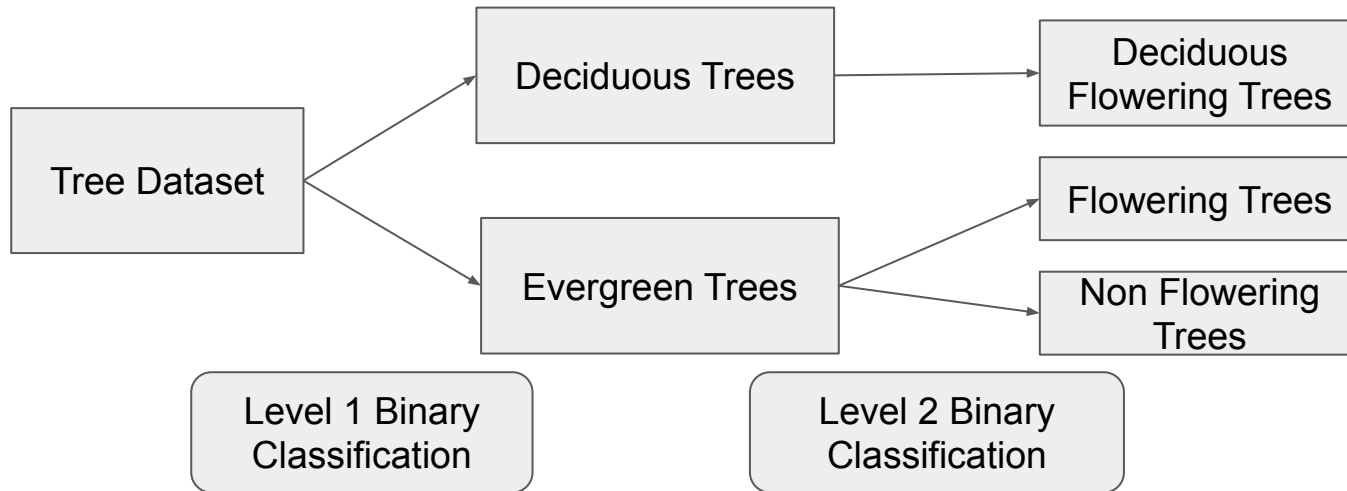
By calculating the average NDVI value for each tree canopy bounding box, we apply a data filter by removing bounding boxes with an NDVI value below 0.25.

After applying the filter, we have 100 genera and a total of 11,628 trees.

Data Processing

Grouping the data: Based on the literature review and our own testing in classifying tree genus, we identified that the imbalanced number of various tree types has impacted the accuracy of our classification results. To address this, our group implemented a Two-level Binary Classification approach.

Two-level Binary Classification Structure:



*Specially, deciduous trees are almost always flowering in biological classification.

Data Augmentation

Purpose: In our dataset, which focuses on the study area of Los Angeles, we faced a significant imbalance between the number of TIFF images representing deciduous trees and those representing evergreen trees. To address this bias and improve the performance of our model, we applied data augmentation techniques. Specifically, we increased the dataset size of the deciduous trees from 2,100 to 8,400 images, creating additional variations of the existing data to balance the representation of both tree types. For level 2 classification, we increased the dataset of the evergreen flowering trees from 2300 to 7000 images. This ensures a more equitable training process for our machine learning model.

Steps of Data Augmentation

1. **Load Data**
2. **Check Image Bands:** The images are checked to ensure they have 4 bands (channels), as the augmentation process specifically handles 4-band images
3. **Apply Data Augmentation transformation patterns:**

```
# Data augmentation transformations
datagen = ImageDataGenerator(
    rotation_range=10,      # Rotate images by up to 10 degrees
    width_shift_range=0.1,  # Shift images horizontally by 10% of the width
    height_shift_range=0.1, # Shift images vertically by 10% of the height
    zoom_range=0.1,        # Randomly zoom the images
    horizontal_flip=True,   # Randomly flip images horizontally
    fill_mode='nearest'    # Filling pixels after transformation
)
```

4. **Save the Output**

Steps of Data Transformation

1. **Directory Traversal:** The code first goes through each subfolder within the source folder. Each subfolder represents a category or label corresponding to different tree species.
2. **Image Loading and Resizing:** For each TIFF image in these subfolders, the code opens the image and resizes it to a fixed size of 128x128 pixels. This standardization ensures that all images have the same dimensions for consistent input into the model.
3. **Normalization:** After resizing, the image is converted to a NumPy array, and its pixel values are normalized to the range [0, 1] by dividing by 255.0. This normalization step is important for machine learning models as it ensures the input values are on a common scale, improving model performance.
4. **Flattening:** The 2D array representing the image is then flattened into a 1D array. This transformation is necessary for feeding the image data into traditional machine learning algorithms, like Random Forest, which expects a flat vector as input.
5. **Label Assignment:** The folder name, which corresponds to the tree species, is used as the label for the image. The code associates each image with its corresponding category, enabling supervised learning.
6. **Storage:** Finally, the flattened image arrays along with their labels are appended to the `data_with_labels_RandomForest` list, preparing the data for further model training.

*About 17574 tiffs has been transformed and stored into the list

Random Forest Model

The model was trained on a dataset of tree species using 75% of the data for training and 25% for testing. The features consist of flattened and normalized image data from TIFF files, and the labels correspond to the tree categories. We used 100 estimators (decision trees) in the Random Forest, and after training the model, it was evaluated using a test set. Performance metrics like **accuracy** and a **classification report** were computed to assess the model's effectiveness.

Results

Level 1 Classification

	precision	recall	f1-score	support
Deciduous	0.97	0.76	0.85	2111
Evergreen	0.82	0.98	0.89	2276
accuracy			0.87	4387
macro avg	0.89	0.87	0.87	4387
weighted avg	0.89	0.87	0.87	4387
Accuracy:	0.8746			

Cross Validation

Cross-Validation Accuracy (5-fold): [0.86854103 0.88069909 0.87879939 0.88107903 0.875]
Mean Cross-Validation Accuracy: 0.8768
Standard Deviation of Cross-Validation Accuracy: 0.0047

Results

Level 2 Classification

	precision	recall	f1-score	support
Evergreen_Flowering_Trees	0.92	0.73	0.81	1794
Evergreen_non-flowering_Trees	0.75	0.92	0.83	1536
accuracy			0.82	3330
macro avg	0.83	0.83	0.82	3330
weighted avg	0.84	0.82	0.82	3330

Accuracy: 0.8204

Cross Validation

Cross-Validation Accuracy (5-fold): [0.82332332 0.81681682 0.84076114 0.81872809 0.81872809]
Mean Cross-Validation Accuracy: 0.8237
Standard Deviation of Cross-Validation Accuracy: 0.0088